

Economics

Higher level

Paper 3

SME Predicted Paper – Hard

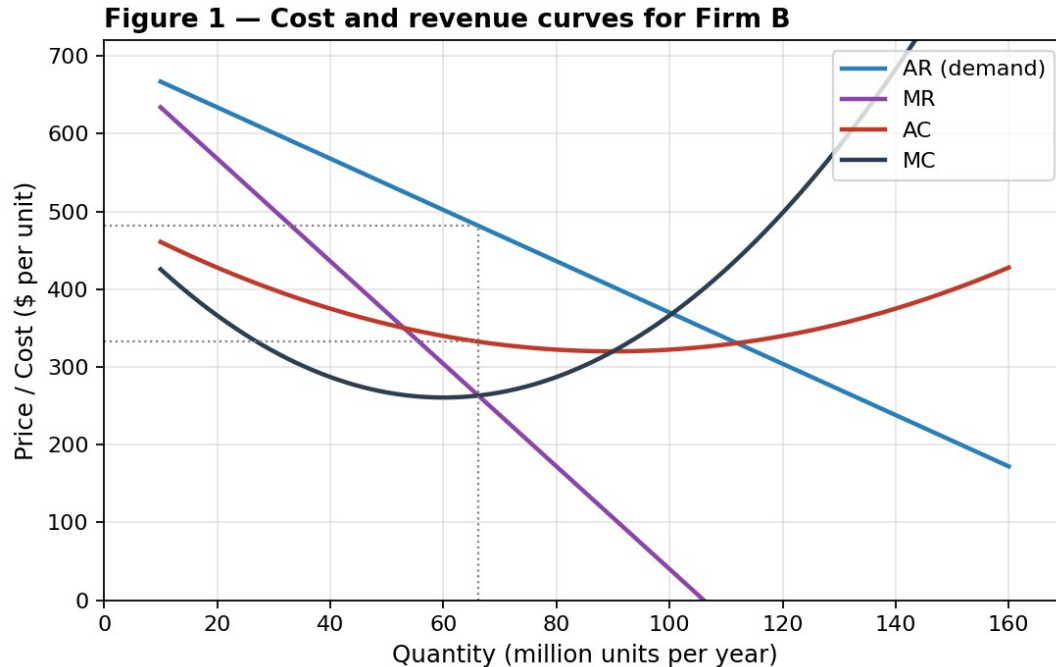
1 hour

Instructions to candidates

- You are permitted access to a calculator for this paper.
- Do not open this examination paper until instructed to do so.
- Answer all questions.
- Answers must be written within the answer boxes provided.
- Unless otherwise stated in the question, all numerical answers must be given exactly or correct to two decimal places.
- You must show all your working.
- Use fully labelled diagrams where appropriate.
- The maximum mark for this examination paper is [60 marks].

1. The global smartphone market is dominated by a small number of firms, including Apple, Samsung, Xiaomi and Huawei. In 2022, the top five firms controlled approximately 72% of global smartphone sales.

Firm B is a typical firm operating in this market. **Figure 1** shows Firm B's demand (AR), marginal revenue (MR), average cost (AC) and marginal cost (MC) curves.



(a) (i) Define the term **concentration ratio**.

[2]

(ii) Using **Figure 1**, identify Firm B's profit maximising level of output.

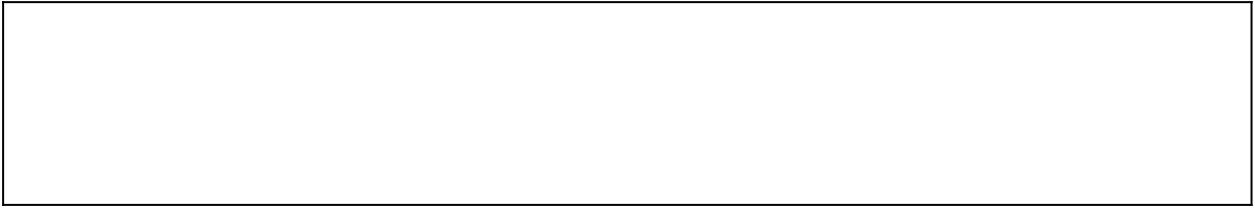
[1]

(iii) Using **Figure 1**, calculate Firm B's short-run economic profit/loss at the level of output identified in part (a)(ii).

[2]

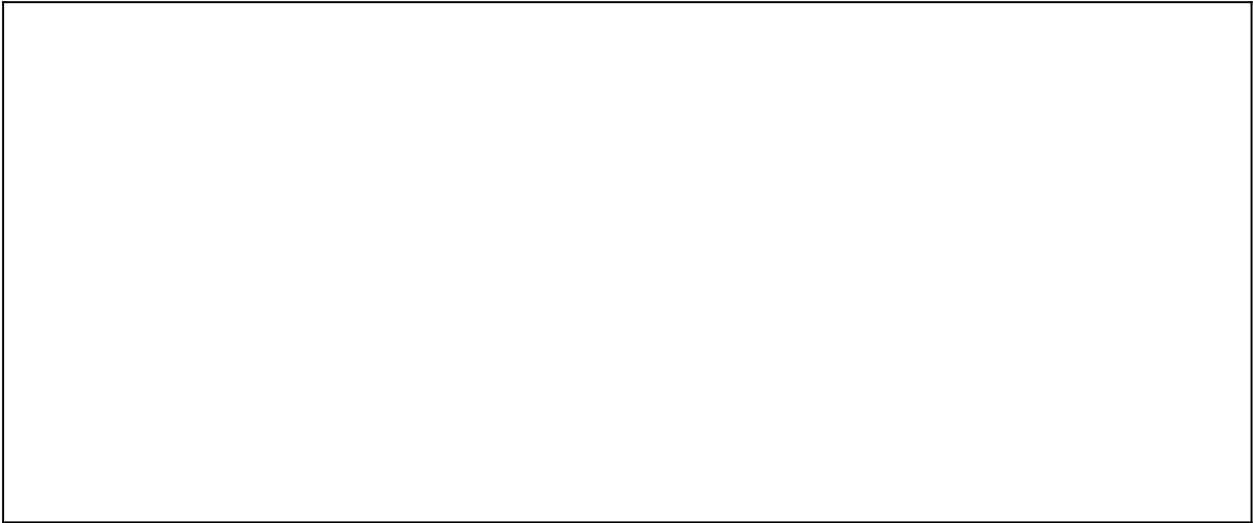
(iv) With reference to **Figure 1**, identify the price charged by Firm B and explain why this firm is able to earn abnormal profit in the long run.

[2]



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(v) Using a kinked demand curve diagram, explain why prices may remain rigid in a non-collusive oligopoly. [4]



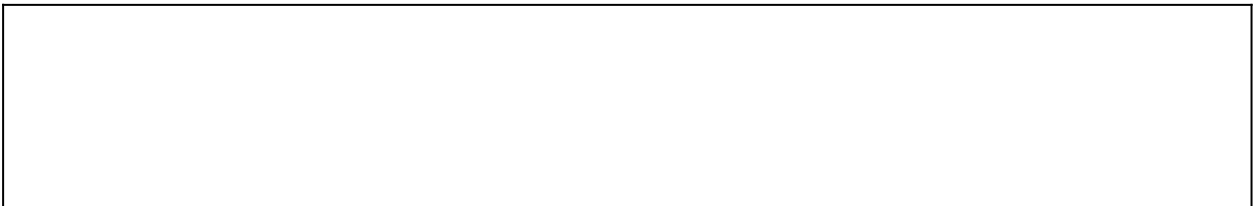
Smartphones rely on rare earth minerals extracted under conditions often associated with significant environmental damage. Many of these minerals are sourced from common pool resources such as shared watersheds and forested land. It is estimated that producing one smartphone generates around 55 kg of CO₂e emissions.

(vi) State **one** of the characteristics of common pool (common-access) resources. [1]



Still, the smartphone industry continues to expand to meet the high global demand. If Firm B produces 66 million units in a year:

(vii) Using the information above, calculate the total CO₂e emissions (in million tonnes) associated with Firm B's output. [2]



Overall, the smartphone industry faces sustainability challenges:

- Rapid product cycles and planned obsolescence promote overconsumption and generate large quantities of electronic waste.
- Mining of rare earth elements can contaminate water sources, harm local ecosystems and pose health risks to communities.
- The industry has a significant carbon footprint; energy-intensive manufacturing and long global supply chains drive greenhouse-gas emissions.

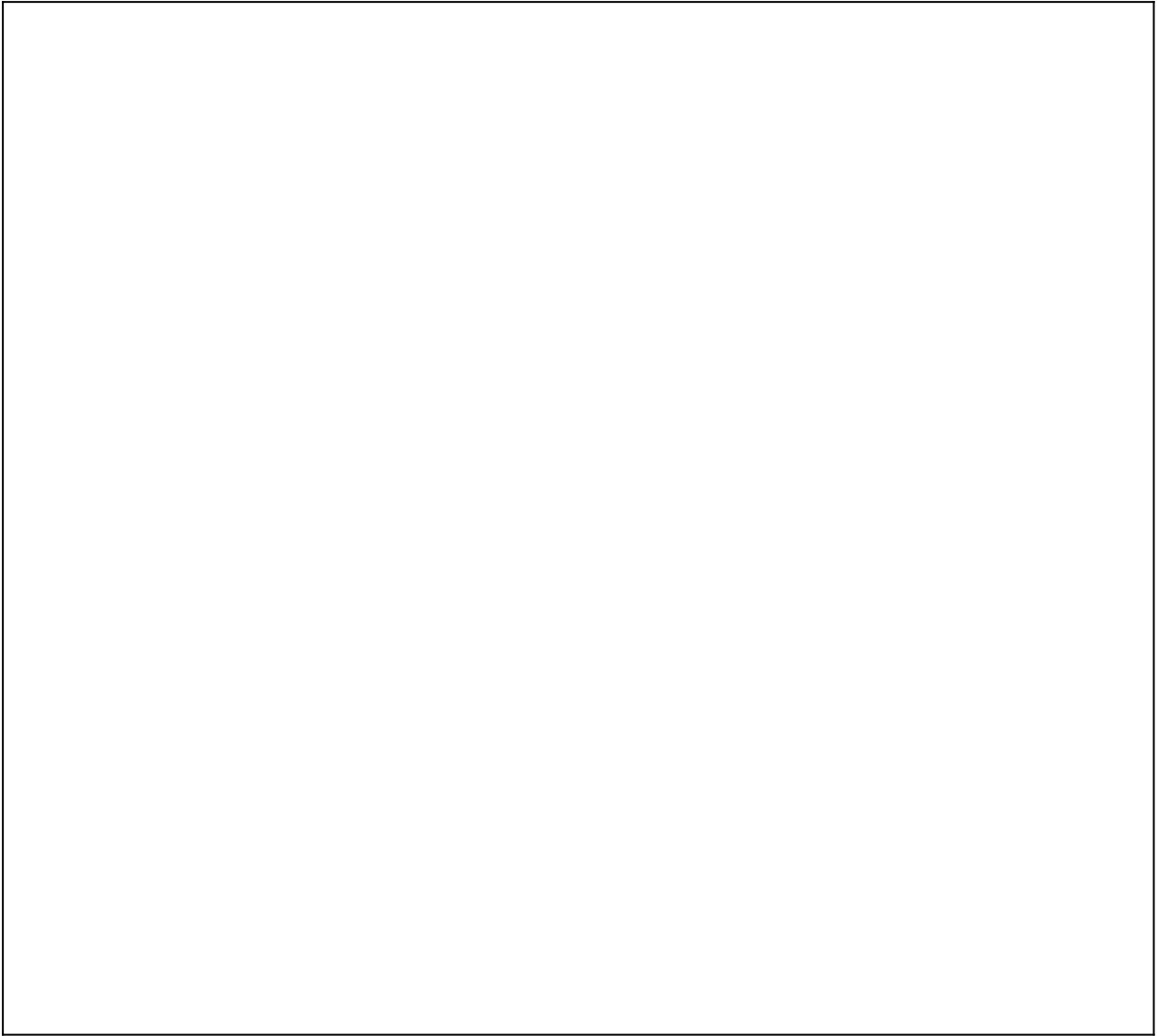
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(viii) Using an externalities diagram and the information provided, explain how smartphone production is responsible for the generation of negative externalities. [4]

In response to environmental concerns, the government of a major economy introduces a specific tax of \$40 per smartphone. Assume that Firm B continues to produce 66 million units in a year.

(ix) Using the information above, calculate the total indirect tax revenue collected from Firm B's output in a year. [2]

(b) Using the text/data provided and your knowledge of economics, recommend a policy that could be introduced to correct the market failures associated with smartphone production. [10]



2. South Africa is an upper-middle-income country and the second-largest economy in sub-Saharan Africa. Table 1 below provides selected macroeconomic data for South Africa.

	Nominal GDP (bn R)	GDP Deflator	Real GDP (bn R)	CPI	Unemployment (%)
2021	6,589	110.4	5,969	108.0	34.3
2022	7,245	118.5		116.4	32.7

Table 1: Selected macroeconomic data for South Africa, 2021–2022

(a) (i) Using the information in **Table 1**, calculate the real GDP for South Africa in 2022 (to 2 dp). [2]

(ii) Using the information in **Table 1**, calculate the rate of inflation in South Africa between 2021 and 2022 (to 2 dp). [2]

In 2022, South Africa's unemployment rate stood at 32.7%, among the highest in the world. Youth unemployment exceeded 60%.

(iii) Define the term **cyclical unemployment**. [2]

(iv) Outline **one** limitation of the official unemployment rate as a measure of labour underutilisation. [2]

South Africa's central bank (SARB) responded to inflation above its 3–6% target band by raising the policy (repo) rate from 3.5% at the start of 2022 to 7.0% by early 2023, with further hikes expected.

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(v) Using an AD/AS diagram, explain how a rise in interest rates is expected to affect real GDP and the price level in South Africa. [4]

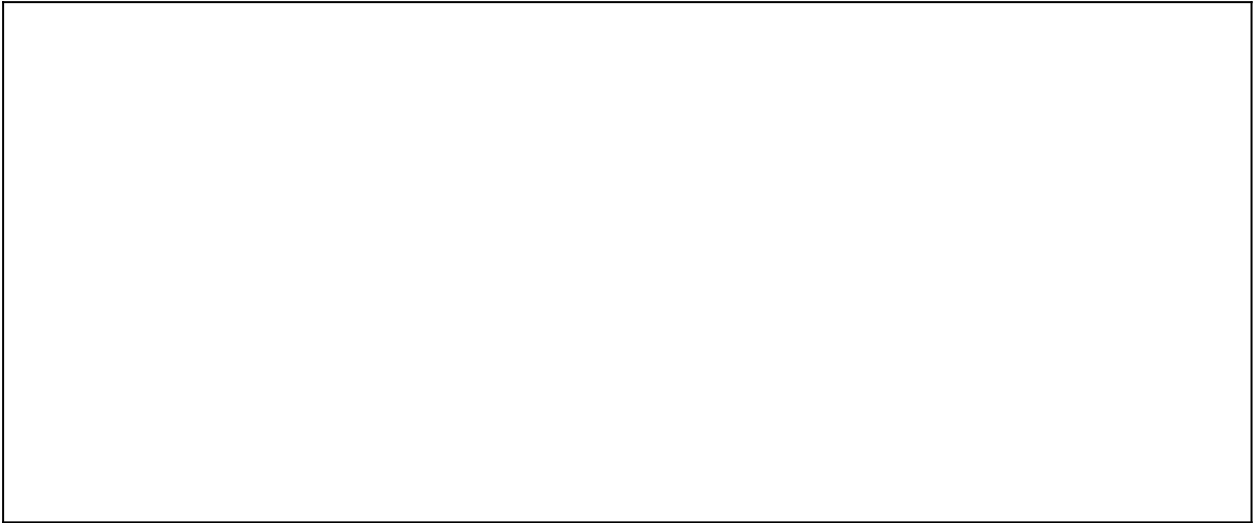
Table 2 below provides selected balance-of-payments data for South Africa in 2022 (US\$ billion).

Item	\$ billion
Exports of goods	123.4
Imports of goods	117.8
Exports of services	14.1
Imports of services	16.2
Net primary income	-12.8
Net secondary income	-2.3

Table 2: Balance-of-payments data for South Africa, 2022

(vi) Using **Table 2**, calculate the current account balance for South Africa in 2022. [2]

(vii) Explain **two** implications of a rising current account deficit. [4]



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Power outages ('load-shedding') cost the South African economy an estimated R560 billion in 2022, reducing business productivity and export competitiveness. Investment in electricity generation has been insufficient, and growth has slowed from around 3% in the early 2010s to below 1% in recent years, highlighting the vulnerabilities of an economy with persistent infrastructure bottlenecks.

(viii) Outline why weak infrastructure can pose a barrier to economic growth and/or economic development. [2]

(b) Using the text/data provided and your knowledge of economics, recommend a policy that could be introduced to revitalize economic growth in South Africa. [10]